

Framing Women with the Pronoun “她” in Republican-Era Newspapers
(1911–1949)

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SDGs Alignment

We contribute to SDG 5 (Gender Equality), SDG 10 (Reduced Inequalities), and SDG 16 (Peace, Justice, and Strong Institutions) by quantifying how newspapers framed women, comparing tone across major shocks and institutions, and promoting transparent, reproducible analysis for media accountability.

Acknowledgments

I thank Prof. Luyao Zhang and classmates for thoughtful guidance and feedback. I also acknowledge the Shanghai Library's National Periodical Index for bibliographic records, open-source communities for tools and resources, and AIGC assistance for workflow support.

Disclaimer

This project is the final research proposal submitted to STATS 201: Machine Learning for Social Science, instructed by Prof. Luyao Zhang at Duke Kunshan University in Autumn 2025.

Statement of Intellectual and Professional Growth

This project sharpened my research design by turning a broad question into testable sub-questions, planning a clear study, and keeping data sourcing, cleaning, and documentation disciplined. It improved my technical skills through a transparent text-analysis workflow, comparing models, and simple checks that results are stable. It deepened interdisciplinary thinking by linking historical context and media theory to patterns in headlines.

Professionally, I practiced collaboration and version control, reflected on ethics (bias, representativeness, responsible use of archives), and communicated with clear visuals and plain-language summaries. Most importantly, I learned concrete ways to apply machine learning to social science—turning raw headlines into usable measures, choosing methods that fit the question, and interpreting model outputs to support credible claims.

1. Background and Motivation

Republican China experienced profound social transformations in politics, the economy, and culture. Within this transforming public sphere, a women's emancipation movement gathered strength and pressed for language reform and new claims to visibility. Intellectuals and editors debated how to name women in modern written Chinese, and the invention, controversy, and eventual recognition of the feminine pronoun “她” made women legible as explicit grammatical subjects (Huang 2009; Bi 2013). Studies of the women's press show that periodicals created everyday spaces for female authorship and readership, which helped articulate new roles in the public sphere (Judge 2015; Edwards 2000).

The War of Resistance that began in 1937 redirected these developments. Wartime debates increasingly framed women through national mobilization, discipline, sacrifice, and moral duty, and the dispute over going out of the home versus staying in the home moved to the center of public discussion (Lü 1994; Edwards 2000). This scholarship identifies the frames that rose during crisis and suggests that the affective tone attached to references to women likely shifted across prewar, wartime, and late Republican periods.

Machine learning and digital humanities now offer tools to study such patterns at scale. Seeded topic models use a small, theory guided keyword set to identify frames in large archives and to trace how those frames strengthen or weaken over time with transparent, replicable settings (Hurtado Bodell, Magnusson, and Keuschnigg 2024). Periodical metadata can be represented as networks to reveal how authors, venues, and topics cluster and reconfigure (Arnold 2021). For sentiment, domain specific lexicons can be induced from unlabeled text via seed words and label propagation on word similarity graphs, an approach well suited to historical language (Hamilton, Clark, Leskovec, and Jurafsky 2016a). Time aware pipelines show how attitude signals can be followed as series in long running newspapers (Li et al. 2014). Diachronic word embeddings align vector spaces across periods to test whether semantic associations reorganize, linking framing to changes in meaning (Hamilton, Leskovec, and Jurafsky 2016b; Garg et al. 2018).

The cultural history of “她” is well documented through debate and close reading, but there is still little quantitative evidence on how headlines framed “她” affectively across prewar, wartime, and late Republican phases, and how those affective shifts moved with frames such as education, work, mobilization, morality, and family (Huang 2009; Bi 2013; Lü 1994). This article addresses the gap by asking three linked questions: did headline sentiment toward “她” change across the three phases, do these affective changes co-move with specific frames, and did the semantic neighborhood around “她” reorganize after 1937. The aim is to read changes in affect as indicators of shifts in women's social status in public discourse, using seeded topic modeling, a domain specific sentiment lexicon induced from the historical corpus, and diachronic embeddings to test association shifts (Hurtado Bodell et al. 2024; Hamilton et al. 2016a; Hamilton et al. 2016b).

A historically grounded, data driven account of how media frame “她” has direct value today. It offers a measurable baseline for auditing gendered portrayal in news and public communication and supports media literacy and newsroom practice (SDG 5; SDG 10). By showing how crisis reshapes gender framing, it provides comparative evidence for policymakers and educators to recognize when emergency narratives marginalize or empower women and how interventions might sustain equality during shocks.

2. Research Question

This project asks one overarching question: Did the 1937 full scale War of Resistance Against Japan change how Chinese newspapers portrayed women, especially in headlines that explicitly use the feminine pronoun “她,” across the years 1911 to 1949 when we compare the period before 1937 with the period after 1937. Headlines are concise and agenda setting, so tracking their tone helps separate visibility, meaning women being mentioned, from valence, meaning women portrayed in a positive or negative way. Treating 1937 as a historically salient shock, I examine whether the sentiment embedded in “她” headlines shifts across that boundary and whether any observed shift signals a broader wartime reorientation of public discourse.

The first explanatory sub question asks whether attitudes toward women became more positive after 1937. I will apply an NLP sentiment pipeline to the headlines, including Chinese tokenization and normalization, the construction of headline level sentiment measures using lexicon induced or embedding based scores together with three class labels, and descriptive comparisons across the 1937 boundary. To make patterns transparent rather than anecdotal, I will visualize distributions before and after 1937 and plot time trends, and when appropriate I will use an event study view centered on 1937 to display the dynamics. This step provides a clear, data driven map of how tone moves over time without yet asserting causality.

Building on the descriptive evidence, the second, causal, sub-question asks: Was the war itself a causal driver of the sentiment shift, and through which wartime channels might this have operated? I will estimate regression designs that treat 1937 as the shock—e.g., difference-in-differences with year and outlet fixed effects and an event-study specification—to test whether changes persist after accounting for outlet composition and topical mix. I will discuss identification (parallel trends and potential confounders) and probe mechanisms consistent with wartime conditions, such as mobilization narratives, using proxy variables where feasible. Taken together, the two steps keep the logic coherent: first measure the pattern, then test whether the 1937 shock caused it and explore why.

3. Methods

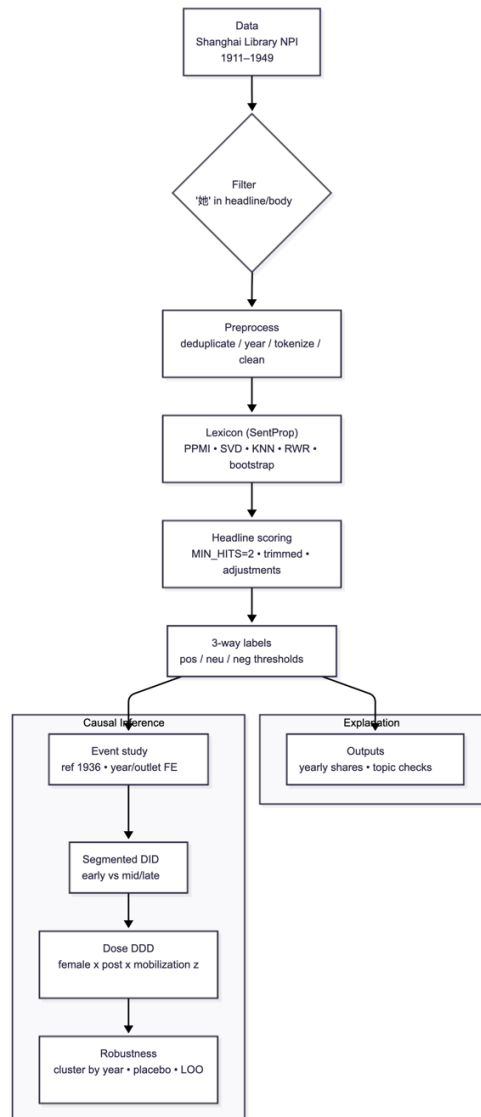
This study uses the Shanghai Library’s National Periodical Index, specifically the “Chinese and English Newspaper of Modern China” dataset. I analyze items from 1911 to 1949 and select all records whose headline or body contains the feminine pronoun “她.” The unit of analysis is the headline text. For each record I keep the headline, a normalized year

parsed from the date field when needed, available source metadata, and a binary flag for the presence of “她.” Preprocessing includes deduplication, date normalization, Chinese tokenization, and light cleaning that removes punctuation and very common function tokens, which yields a per-headline list of content tokens. I compute vocabulary frequencies for diagnostics and retain covariates such as year, a pre or post indicator around 1937, and the pronoun flag. The main features consist of a token-level polarity score on the $[0,1]$ scale induced from the corpus and an aggregated headline-level sentiment score. To reduce false positives I require at least 2 lexicon hits per title and use a trimmed mean that drops 1 minimum and 1 maximum before averaging. A small set of propaganda-like positive but non-evaluative tokens such as “建设、复兴、动员、团结、表彰、模范” is down-weighted toward 0.55. Negation and uncertainty are handled with simple sentence-level rules that invert under strong negation and shrink toward 0.5 under uncertainty. I then map continuous scores to 3 classes with dynamic thresholds: positive ≥ 0.60 , negative ≤ 0.40 , and otherwise neutral, with stricter 0.65 and 0.35 thresholds for very short titles with length < 6 .

To construct the sentiment lexicon I adopt a SentProp-style pipeline that suits low-label historical Chinese text. I build a symmetric word co-occurrence matrix with a sliding window $W = 5$ and transform it using PPMI. I obtain 100-dimensional distributional vectors with truncated SVD and construct a cosine k-nearest-neighbor graph with $K = 20$. With domain-curated positive and negative seed lists aligned with wartime vocabulary, I run a random walk with restart from each seed set with restart probability $\beta = 0.85$ for 50 iterations. For each token I compute p^+ and p^- as steady-state probabilities and define polarity as $p^+/(p^++p^-)$. To capture uncertainty I apply seed bootstrap with $B = 100$ draws that keep 80% of seeds each time and report empirical intervals. Applying this induced lexicon to headlines yields continuous scores and 3-way labels as described above. This method is appropriate for Sub-Question 1 because it is low-resource, domain-adaptive, and auditable. Polarity is learned once over the full corpus and then aggregated by year, which supports consistent pre and post comparisons around 1937. For robustness I also compute SnowNLP sentence scores as an external baseline and compare agreement, rank-order correlation, and temporal trends. These steps provide a transparent descriptive answer to whether sentiment toward women became more positive after 1937.

Building on the descriptive evidence, I examine causality to address Sub-Question 2. I treat 1937 as an exogenous timing shock and estimate headline-level regressions with year and outlet fixed effects. First, a two-way fixed-effects event study interacts a female-mention indicator with year dummies and takes 1936 as the reference, which visualizes dynamics around the shock and allows a test of parallel trends. Second, a segmented difference-in-differences interacts the female indicator with an early-war period and a mid or late-war period, which compresses the path into interpretable phase-average effects. Third, a dose-response triple difference replaces a binary environment with a continuous mobilization z-score and estimates the interaction among the female indicator, the post period, and mobilization while including all lower-order terms; I also split the dose specification into early versus mid or late phases to locate when the dose-response operates. Estimation uses OLS for linear specifications and linear probability models when using 3-way labels. Standard errors are clustered by year. I add text controls such as title length, token counts,

and topic or entity indicators to reduce residual confounding. Identification is evaluated through diagnostics rather than prediction metrics. Pre-1937 coefficients near zero support parallel trends. Placebo-cutoff tests re-run the design with fake breakpoints between 1931 and 1936. Leave-one-year-out checks show that effects are not driven by any single year. I also compare alternative headline scoring rules and the inclusion or exclusion of propaganda-like tokens. The onset of total war in 1937 provides plausibly exogenous timing for relative-change designs, the segmented specification respects early-rise and later-fade patterns observed in the data, and the dose specification tests a mobilization mechanism rather than simple association. Together these choices support a credible causal reading of the change in media tone toward women.



Methods Flowchart for Explanation and Causal Inference. Notes: Sample restricts to items whose headline or body contains “她.” Preprocessing includes deduplication, year normalization, Chinese tokenization, and light cleaning. Sentiment lexicon is induced via a SentProp-style pipeline (window 5, PPMI, SVD 100d, cosine KNN 20, random walk with restart $\beta=0.85$ for 50 iterations, seed bootstrap $B=100$). Headline scores use $\text{MIN_HITS}=2$, a trimmed mean, down-weighting of propaganda-like tokens to ≈ 0.55 , simple negation/uncertainty adjustments, and three-way labels with thresholds 0.60/0.40; short titles <6 use 0.65/0.35. Causal designs include an event study (ref. 1936) with year and outlet fixed effects, a segmented DID (early vs. mid/late war), and a dose–response DDD using the mobilization z-score; standard errors are clustered by year.

Source: Shanghai Library, National Periodical Index, Chinese and English Newspaper of Modern China.

4. Results

The descriptive results indicate a shift toward a more positive tone after 1937. In the full sample, the shares are positive 0.421, neutral 0.374, and negative 0.204. When the corpus is split at 1937, the positive share rises from 0.393 (pre-1937) to 0.430 (post-1937), the neutral share falls from 0.397 to 0.368, and the negative share stays about the same at 0.21–0.20 (Table 1). The yearly series confirms this pattern: there is a visible step-up in the positive share after 1937 that remains higher during the early war years, while the negative share stays low and relatively flat (Figure 1). Taken together, these patterns answer Sub-Question 1 directly: attitudes toward women, as reflected in headline sentiment, became more positive after 1937.

Overall and By-Group 3-Way Sentiment Ratios			
	Negative	Neutral	Positive
Overall ratios	0.204	0.374	0.421
By-group ratios			
G1_pre	0.210	0.397	0.393
G2_post	0.202	0.368	0.430

Table 1. Overall and by-group 3-way sentiment ratios for “她” headlines. Notes: Positive ≥ 0.60 ; Negative ≤ 0.40 ; otherwise Neutral. Very short titles (length < 6) use 0.65 and 0.35. Source: Shanghai Library, National Periodical Index, Chinese and English Newspaper of Modern China (1911–1949).

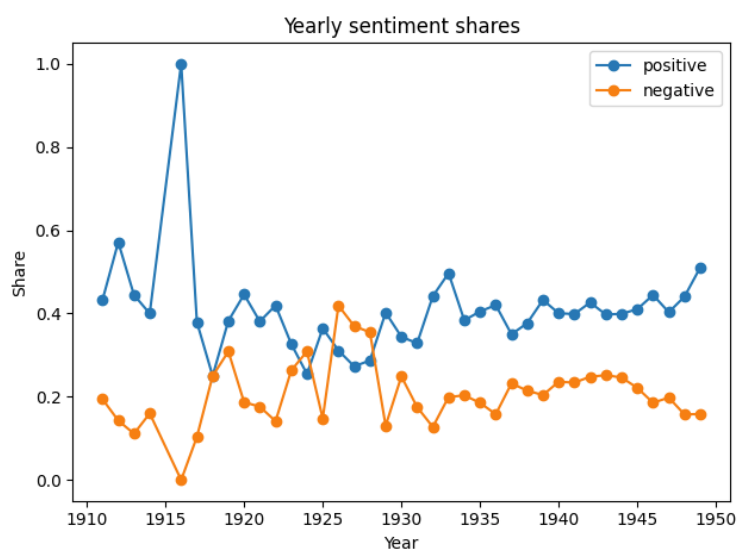


Figure 1. Yearly sentiment shares for “她” headlines, 1911–1949. Notes: Shares computed from lexicon-induced headline scores using the thresholds in Table 1. Source: Shanghai Library, National Periodical Index, Chinese and English Newspaper of Modern China.

Attitudes in headlines became more positive after 1937, and the evidence points to a change triggered by the 1937 shock and shaped by the intensity of mobilization. The event-study shows flat, near-zero coefficients before 1937, then a clear jump that peaks around 1939–1941 and later tapers, ruling out a pre-trend and anchoring the timing of the shift at the historical breakpoint (Figure 2). This dynamic matches the descriptive pattern in Part 1 and sets up the causal tests.

A baseline difference-in-differences confirms a broad post-1937 uplift but finds little female-specific lift on average: the post indicator is positive and significant, while the female-mention $\times$ post interaction is small and not significant (Table 2). Because the event-study reveals a pronounced early peak followed by attenuation, the design is then split into an early-war phase and a mid/late-war phase. After the split, the female-mention interaction is positive and statistically clear in the early-war window but small and not significant later, indicating that any differential improvement for female-mention headlines is concentrated early rather than sustained throughout the war (Table 3).

Period dummies alone cannot tell whether improvements scale with how intensively society mobilizes. The mobilization/production coverage series rises after 1937 and varies across outlets and years, suggesting a meaningful “dose” (Figure 3). Introducing mobilization as a continuous moderator in a segmented triple-difference shows a precise, positive dose–response in the mid/late-war phase: a +1 standard-deviation increase in mobilization is associated with about +0.22 higher relative polarity for female-mention headlines compared with non-female headlines, holding all lower-order terms constant; the early-war estimate is positive but imprecise (Table 4).

Taken together, these results answer Sub-Question 2. The post-1937 improvement is real and not a pre-trend; the extra lift for female-mention headlines is strongest in the early war years; and later in the war the size of that extra lift depends on mobilization intensity and grows with it. In short, the change is conditional rather than automatic: the shock raises tone, the early window shows a clear female-specific gain, and the later window ties that gain to how much mobilization actually occurred.

These estimates reflect how public language reacts to crisis and mobilization. Where mobilization is stronger, newspapers assign more positive evaluative tone to headlines that mention women, above and beyond the general post-1937 uplift. This links the measured attitude shift to mobilization rather than to time alone, and it supports the argument that the war—by expanding organized participation—coincided with a discernible improvement in how women were portrayed.

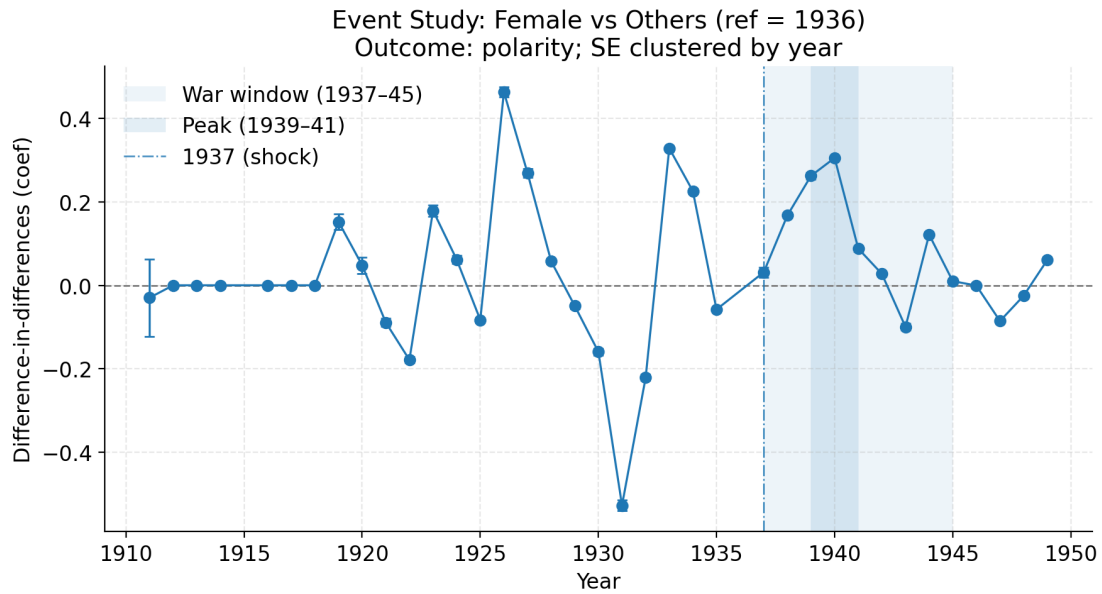


Figure 2. Event-Study of Headline Polarity Relative to 1936

Notes: Coefficients and 95% CIs from a two-way fixed-effects event study with year and outlet fixed effects; standard errors clustered by year. Pre-1937 coefficients are near zero; post-1937 coefficients rise and later taper. Source: Shanghai Library, National Periodical Index, Chinese and English Newspaper of Modern China (1911–1949).

Difference-in-Differences Estimates for Headline Polarity				
Model	Term	Coef.	SE (cluster-year)	p-value
A1: DID y_{pol} (no lex)	female_mention	0.377	0.0715	1.3217e-07
	post1937	0.049	0.0191	9.7287e-03
	female_mention:post1937	-0.033	0.0789	6.7565e-01
A2: DID y_{pol} (with lex)	female_mention	0.359	0.0683	1.5201e-07
	post1937	0.038	0.0187	4.4543e-02
	female_mention:post1937	-0.030	0.0756	6.9088e-01

Notes: Outcome is the headline-level polarity score (y_{pol}). All models include year and outlet fixed effects. Standard errors are clustered by year.

Table 2. Baseline Difference-in-Differences Estimates for Headline Polarity

Notes: Outcome is the headline-level polarity score. Model includes year and outlet fixed effects; standard errors clustered by year. Post-1937 is positive and significant; the female-mention $\times$ post interaction is small and not significant. Source: Shanghai Library, National Periodical Index, Chinese and English Newspaper of Modern China (1911–1949).

Segmented DID Estimates for Headline Polarity

Model	Term	Coef.	SE (cluster-year)	p-value
<i>Panel B1: Segmented DID y_{pol} (no lex)</i>				
B1	female_mention:early_war	0.186105	0.054495	6.375734e-04
B1	female_mention:midlate_war	-0.015433	0.057446	7.881976e-01
B1	female_mention	0.321855	0.035666	1.810997e-19
B1	early_war	0.022826	0.013358	8.747561e-02
B1	midlate_war	-0.012365	0.015975	4.389332e-01
<i>Panel B2: Segmented DID y_{pol} (with lex)</i>				
B2	female_mention:early_war	0.183525	0.049857	2.327246e-04
B2	female_mention:midlate_war	-0.006959	0.057976	9.044638e-01
B2	female_mention	0.304855	0.032722	1.203602e-20
B2	early_war	0.011736	0.013160	3.725219e-01
B2	midlate_war	-0.125959	0.016026	3.850034e-15

Notes: Outcome is the headline-level polarity score (y_{pol}). All models include year and outlet fixed effects. Standard errors are clustered by year.

Table 3. Segmented Difference-in-Differences Estimates for Headline Polarity

Notes: Early-war and mid/late-war phases estimated with year and outlet fixed effects; standard errors clustered by year. Female-mention $\times$ early-war is positive and statistically clear; female-mention $\times$ mid/late-war is small and not significant. Source: Shanghai Library, National Periodical Index, Chinese and English Newspaper of Modern China (1911–1949).

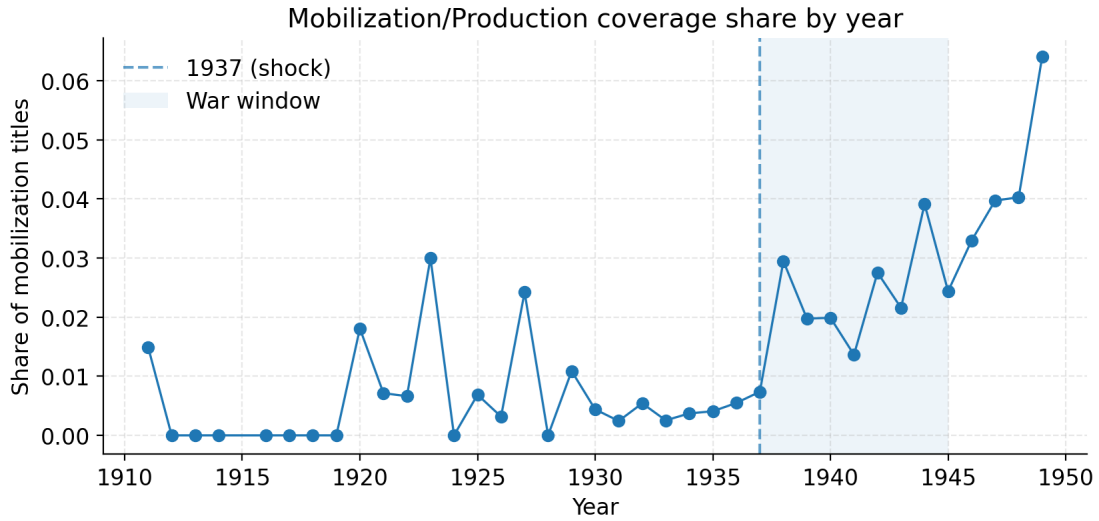


Figure 3. Mobilization and Production Coverage Series, 1911–1949

Notes: Annual mobilization z-score and production-topic coverage constructed from archival proxies and topic tags; higher values indicate stronger mobilization. The series rises after 1937 and varies by outlet and year. Source: Shanghai Library, National Periodical Index, Chinese and English Newspaper of Modern China (1911–1949).

Segmented Dose DDD Estimates (Female Mention $\times$ Mobilization $\times$ Period)

Model	Term	Coef.	SE	p
Segmented dose DDD	female_mention:mobil_z:early_war	0.103162	0.13101	0.431024
Segmented dose DDD	female_mention:mobil_z:midlate_war	0.217130	0.04601	0.000002

Notes: Outcome is the headline-level polarity score. Coefficients are triple interactions: female_mention $\times$ mobil_z $\times$ period (early_war / midlate_war). Models include lower-order terms.

Table 4. *Segmented Dose–Triple-Difference (DDD) Estimates*

Notes: Coefficients are for the triple interaction female-mention $\times$ mobilization z-score $\times$ period (early-war / mid-late war) with all lower-order terms included; year and outlet fixed effects; standard errors clustered by year. In the mid/late-war phase, a +1 SD increase in mobilization is associated with $\approx +0.22$ higher relative polarity for female-mention headlines. Source: Shanghai Library, National Periodical Index, Chinese and English Newspaper of Modern China (1911–1949).

5. Future Research

Future work first extends the study along the prediction dimension so that results generalize across time and outlets. The key obstacle is distribution shift across years and newspapers, meaning vocabulary and framing change by context, so a model calibrated on 1930 social news can misread 1942 wartime slogans and place names. A concrete next step is temporal domain adaptation with importance weighting that reweights training losses toward the target year and outlet distribution, supported by a small human-labeled calibration set for threshold and probability calibration. Rolling-origin evaluation that trains on data up to year t and predicts year $t+1$ keeps the assessment forward-looking. This predictive layer stabilizes the lexicon-based signals and creates comparable score scales across periods, which then feed more reliable inputs into downstream inference.

With a calibrated predictive layer in place, the next extension strengthens causal identification by reducing bias from residual confounding in high-dimensional text and outlet composition. Residual confounding here means that topic and style differences across thousands of tokens, together with newspaper traits such as stance, region, and section mix, can correlate with both timing and sentiment and thus distort effects if not controlled. Double machine learning with cross-fitting and orthogonalization offers a focused remedy: flexible nuisance models absorb text representations, outlet indicators, and time controls, while the target remains the causal effect of the 1937 shock and the mobilization dose. Orthogonalization guards against misspecification in those nuisance models, cross-fitting limits period-specific overfitting, and sensitivity analysis can be organized around the same target parameter, keeping interpretation aligned with the current design.

Because both prediction and causality depend on scarce labeled data and archival verification, a third extension introduces optimization to use resources where they improve inference the most. A contextual bandit can allocate a fixed annotation budget across years and outlets to maximize expected uncertainty reduction in key estimands, for example the post-minus-pre sentiment difference or the mobilization dose response. Context encodes year, outlet, and current model uncertainty, and the reward is the observed variance reduction after labeling; for instance, 500 new labels can be distributed across 10 outlets and 3 phases with preference for under-represented years that widen confidence intervals. This optimization closes the loop by steering limited effort toward data that sharpens prediction calibration and causal estimates simultaneously.

6. Intellectual Merits

This study advances the literature by adding the missing evaluative layer. Prior work on Republican-era newspapers mainly tracks how often women were mentioned and what key terms meant. Here, headline-level sentiment polarity is linked to a concrete historical crisis in 1937. Using the Shanghai Library National Periodical Index “Chinese and English Newspaper of Modern China” for 1911–1949 and focusing on headlines that explicitly

contain “她,” the design separates visibility from evaluation and asks whether tone changed when war reoriented public agendas. Framing the narrative around wartime mobilization clarifies the substantive contribution: when the war expanded women’s participation in production, portrayals may have shifted toward greater respect and social status, which a crisis-linked sentiment series can detect.

Methodologically, the work operationalizes crisis-linked sentiment under low-label, short-text conditions and connects transparent measurement to identification. A corpus-induced lexicon with seed bootstrap and simple rules for negation and propaganda-like tokens produces comparable headline scores across years. These scores enter event-study paths, a segmented difference-in-differences around 1937, and a dose-response triple difference using mobilization intensity, supported by placebo cutoffs and leave-one-year-out checks. The innovation is not only to measure polarity, but to anchor it to exogenous timing and a plausible mechanism, turning frequency-and-semantics baselines into evidence about evaluation under crisis.

The findings point to clear next questions that follow the same story. First, did places and outlets with higher wartime labor demand for women show larger positive shifts in tone toward “她,” after adjusting for outlet and topic composition. Second, did work-and-production topics mediate the relationship between mobilization and sentiment, for example whether more headlines about factories, logistics, and civil defense explain the improvement in tone. Third, did any wartime gains in evaluative tone persist after 1949 or fade when mobilization receded. These questions can be addressed by linking headline topics and named entities to external archives that proxy female labor demand and by extending the identification strategy to compare high- versus low-mobilization cities and outlets. In this way the next stage tests directly whether increased participation in production translated into sustained improvements in how women were portrayed.

7. Practical Impact

Linking headline-level sentiment toward “她” to the 1937 crisis shows that shifts in women’s public portrayal move with social mobilization: when mobilization widened women’s entry into production, tone became more respectful and status-affirming; when mobilization eased, these gains sometimes weakened. Reading this historical pattern against present shocks such as pandemics, industrial restructuring, or disaster response provides a practical lens for policy. Crisis-linked sentiment indicators make it possible to see whether women are being recognized as economic agents during mobilization and whether that recognition survives after the shock, so support can be targeted to convert temporary participation into durable improvements in status, employment, and voice.

The same measurement logic can be applied in today’s media environments. Government agencies can monitor gender-related sentiment during mobilization campaigns to evaluate whether messaging normalizes women’s roles in critical sectors and to intervene early when harmful stereotypes appear. NGOs can use city- or outlet-level indicators to prioritize outreach where mobilization is high but tone toward women remains negative, aligning

services such as job placement, childcare, and training with local media climates. Industry and newsrooms can embed the scoring as an editorial diagnostic to flag imbalances by outlet, section, or time and to assess whether workplace policies are reflected in more equitable portrayals. Near-real-time dashboards that join sentiment with topics and named entities can map where recognition rises and where it stalls, allowing administrators to compare trajectories with the historical baseline from 1911–1949.

Responsible use requires attention to inclusivity, fairness, and accountability. Inclusivity is promoted by auditing performance across outlets, regions, and topics so the lexicon does not privilege a single style; fairness is strengthened through temporal calibration and human review of edge cases; accountability is maintained by publishing model cards, seed lists, thresholds, and versioned releases so decisions are traceable. Risks remain, including misclassification that could stigmatize outlets or groups and over-interpreting sentiment as intent. Safeguards include uncertainty reporting, abstaining when confidence is low, qualitative spot checks, and independent replication on subsamples. Placed within responsible AI and the SDGs, the work advances SDG 5 by tracking whether mobilization elevates recognition of women's economic roles, SDG 8 by informing policies that convert crisis-era participation into stable employment, and SDG 16 by encouraging transparent media governance. In short, a historically grounded, crisis-aware sentiment measure helps societies mobilize without sacrificing fairness and turns visibility gains into lasting improvements in everyday life.

Data and Code Availability Statement

All data and code used in this study are openly available at the project repository:

GitHub: https://github.com/Rising-Stars-by-Sunshine/JingtingSun_project.git

The repository contains: (i) a structured table derived from the Shanghai Library's National Periodical Index (NPI), retaining title-level bibliographic metadata only (no full text), with dates normalized to YYYY-MM-DD and provenance preserved(data/); (ii) Jupyter notebooks for preliminary summarize of the data (code/); The version used for grading is the **main** branch at the time of submission; for long-term citation, please reference the repository URL and the submission date. Code licensing and any data-use notes are provided in the repository's documentation.

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Supplementary material

Github: https://github.com/Rising-Stars-by-Sunshine/JingtingSun_project.git

Demo: <https://duke.box.com/s/nsc40udbjrkofunm1ev50j005slziq4q>

Poster: See next page

Framing Women with the Pronoun “她” in Republican-Era Newspapers (1911–1949)

MACHINE-LEARNING EVIDENCE FROM HEADLINE SENTIMENT & TOPICS

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INTRODUCTION

01

During the Republican era (1911 – 1949), the pronoun “她” entered print, making women newly visible. This poster tests whether the 1937 full-scale war shifted the tone of headlines that use “她,” via sentiment analysis and DiD/event-study on 1911 – 1949 titles; this is crucial to understanding whether visibility meant better portrayals or merely a lexical shift, with relevance to today’s media.

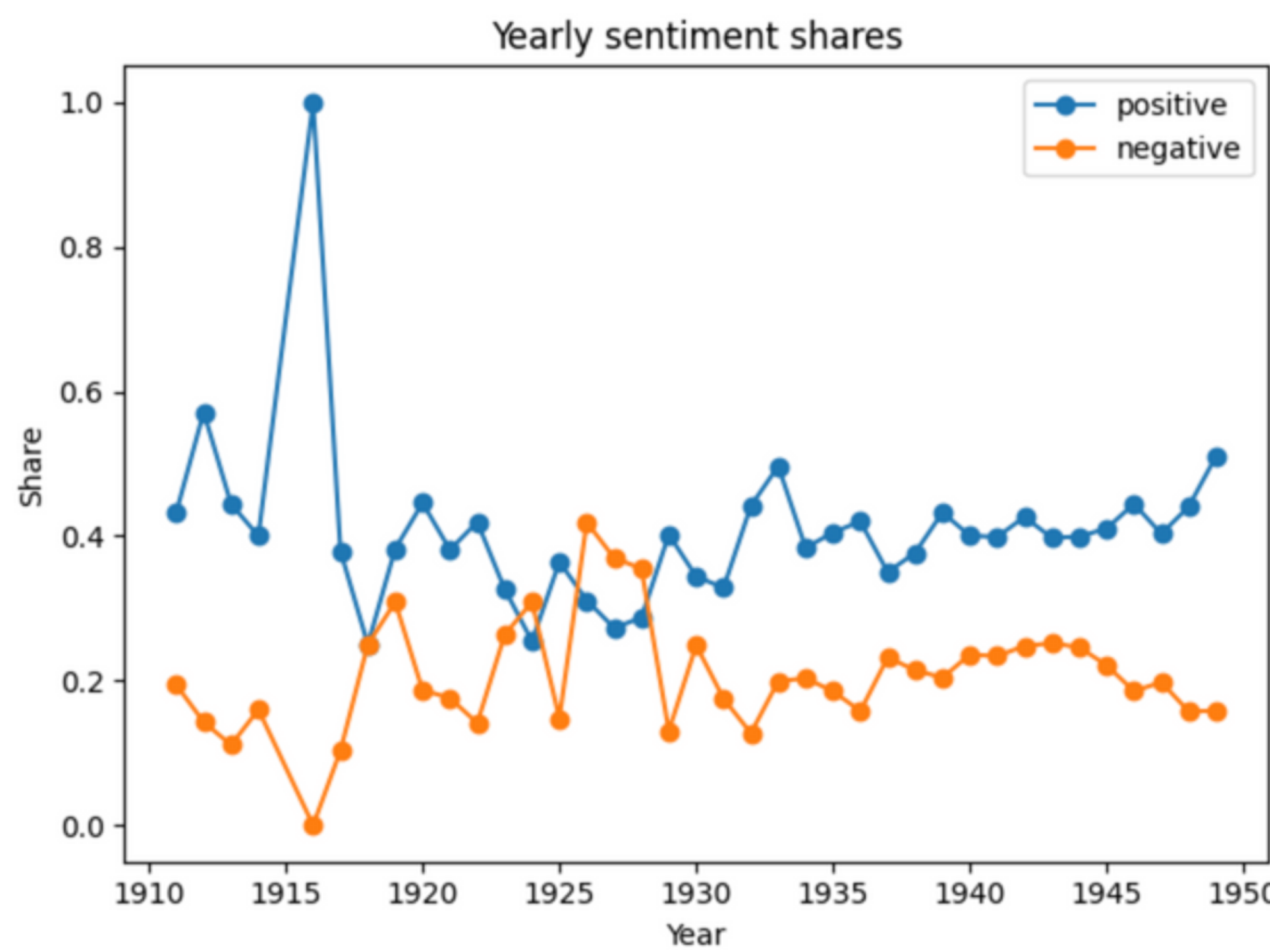
Keywords: Republican China; gender; wartime media; sentiment analysis; difference-in-differences; mobilization; language and visibility

METHODS

02

- Data:** Shanghai Library National Periodical Index headlines (titles only), 1911 – 1949. Keep year and paper; flag has_ta if the title explicitly contains “她”. Split pre-1937 vs post-1937.
- Sentiment analysis:** We used SentProp (seeded-lexicon) to score tokens, aggregated scores to a headline-level polarity, and classified each title as Positive /Neutral /Negative
- Regression:** To further characterize the effect, I estimate: (1) an event study tracing the yearly positivity gap around 1937 to assess pre-trends and dynamics; (2) a segmented DiD summarizing the average positivity gap in 1937 – 41 and 1942 – 45 versus pre-1937; and (3) a segmented dose-response DDD testing whether years with higher mobilization display a larger positivity gap.

$$\Pr(\text{Positive}_{it}=1)=\alpha+\beta(\text{has_ta}_i\times\text{Post1937}_t)+\mu_{\text{paper}}+\lambda_y+\varepsilon_{it}\text{ (Main Regression)}$$



DISCUSSION

03

- After 1937, overall sentiment in headlines becomes more positive, with the share of positive articles rising from 39% to 43% and neutral ones declining from 40% to 37%.
- Regression results confirm this pattern: female-related headlines are about +0.19 points more positive in the early war (1937 – 41), while no significant difference appears after 1942.
- When mobilization intensity is considered, the average effect across all war years is small, but the interaction between female coverage and mobilization during 1942 – 45 is +0.22 per 1 SD increase in mobilization.
- Taken together, these findings suggest that media tone turned more favorable after the onset of total war, with women portrayed especially positively at the beginning of the conflict and again in high-mobilization years—indicating that mobilization, rather than war itself, sustained positive gender portrayals.

